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Studierichting: Informatica
Oefeningenreeks 6B nr. 5.8

$$\frac{s_6}{s_2} = \frac{-84}{-4}$$

$$\Leftrightarrow \frac{x_1 \dfrac{1-q^6}{1-q}}{x_1 \dfrac{1-q^2}{1-q}} = 21$$

$$\Leftrightarrow \frac{(1-q^6)(1-q)}{(1-q^2)(1-q)} = 21$$

$$\Leftrightarrow \frac{1-q^6}{1-q^2} = 21$$

$$\begin{array}{ccccccc|ccc}
-q^6 & +0q^5 & +0q^4 & +0q^3 & +0q^2 & +0q & +1 & -q^2 & +0q & +1 \\ \cline{8-10}
q^6 & -0q^5 & -q^4 & & & & & q^4 & +q^2 & +1 \\ \cline{1-3}
 & & -q^4 & +0q^3 & +0q^2 & & & & & \\
 & & q^4 & -0q^3 & -q^2 & & & & & \\ \cline{3-5}
 & & & & -q^2 & +0q & +1 & & & \\
 & & & & q^2 & -0q & -1 & & & \\ \cline{5-7}
 & & & & & & 0 & & &
\end{array}$$

$$\Leftrightarrow \frac{(-q^2+1)(q^4+q^2+1)}{1-q^2} = 21$$

$$\Leftrightarrow q^4 + q^2 + 1 = 21$$

$$\Leftrightarrow q^4 + q^2 - 20 = 0$$

$$\overset{t=q^2}{\Rightarrow} t^2 + t - 20 = 0$$

$$\Delta = 1^2 - 4(-20) = 81$$

$$\Rightarrow t = \frac{-1-9}{2} = -5 \vee t = \frac{-1+9}{2} = 4$$

$$\Rightarrow q = \pm 2$$

$q = -2$:

$$x_1 \frac{1-(-2)^2}{1-(-2)} = -4 \Leftrightarrow x_1 = \frac{-4 \cdot 3}{-3} \Leftrightarrow x_1 = 4$$

$q = 2$:

$$x_1 \frac{1-2^2}{1-2} = -4 \Leftrightarrow x_1 = \frac{-4 \cdot (-1)}{-3} \Leftrightarrow x_1 = -\frac{4}{3}$$

References